

COMPUTATION REPORT: CONTROLLED REGRET-SCALED CERTIFICATION

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This report identifies the executed objects, arithmetic contracts, registration amendment, and verification boundaries for R43. It accompanies the complete manuscript and technical supplement and does not substitute publication hashes for mathematical or numerical validation. All earlier scientific files remain unchanged.

1. SCIENTIFIC AND REPOSITORY IDENTITY

The addressed report is `reviews/2026-09-25-econometrica-r40/referee_report.md` at commit `cf195a594c177208dbb5488576464dce9627a830`. The revision builds on the published R42 head `7d54e728a9732979605697895e3a07444dc1e528`, not on an earlier manuscript snapshot. The new branch is `revision/econometrica-r43-regret-scaled-atomic-2026-09-25`. No merge into the default branch or mutation of another revision/review branch is part of this publication.

The current four documents use the repository's unchanged Econometrica class and support files. The original scientific content remains in the inherited files. New source is under `revisions/2026-09-25-r43`. The human-readable review index, manifest, and branch preservation audit distinguish navigation changes from new scientific content.

2. REGISTRATION AND AMENDMENT

The initial protocol and three source hashes were committed before evaluation at `18cbcf41a4c35660be824f6984f68a1d38288a9`. Only development seed 430000 had been executed before that registration. The initial run exposed excessive distinct-denominator accumulation, and an integer-string conversion limit. The replacement uses outward dyadic summand enclosures in the lower-certificate residual and upward-rounded theoretical diagnostics. Its hashes were registered at `950b1f1ac2ebaf974d4ef528add8b2c2e4ce72b`. The case parameters and LP proposals were not changed. Initial code and partial results are preserved under `history/initial_frozen`.

This sequence is described as predeclared cases with a disclosed code amendment. It is not an untouched code-frozen holdout. The final audit checks the current sources against the amended hashes and checks every case against the unchanged original registration. The source-hash equality is recorded by filename in `results/publication_rechecks.json`. It establishes identity, not correctness of the analytic theorems.

3. EXECUTED OBJECTS AND EXACT CHECKS

There are 16 controlled finite root computations: eight environments and two relaxations. The fixed-model allowance study has eight computations: four allowances and two relaxations. The controlled continuum example uses 17 finite fibers, from which four partition intervals are formed with exact rational weights. All 41 endpoint objects are independently reconstructed, including time-capped cases. These objects contain 6,648 state/date operating checks in total. A stopped LP can return a sound wide interval; it is not discarded or counted as a solved constrained optimum.

The rational reader reconstructs all primitive dimensions and stochastic rows, the exact finite operating optimum and reference policy, disadvantage and savings coefficients, product bounds,

Bellman equalities, cone inequalities, dual residual lower endpoint, deployed common policy, every original restart allowance, and implementation objective. The separate publication audit independently regenerates every registered primitive draw and recomputes each continuum sum. It does not import the constructor when checking these objects. Four mutation checks reject a forged endpoint, negative policy probability, negative inequality multiplier, and nonstochastic primitive row.

The main controlled suite contains 4 capped LP calls in the recorded execution. A material reduction in interval width occurs in 5 of eight pairs. At the fixed eight-period small allowance, the scaled positive interval width is at most 0.000008657. The 16-cell controlled continuum interval is $[10.074215, 10.184476]$. These are distinct experiments, not replacements for the 30 unresolved positive-cost intervals in the original continuous-condition development cohort.

4. TIMING, MEMORY, AND STOPPING

The LP cap is 30 seconds and belongs only to the numerical proposal. Model/relaxation construction, rational constructor arithmetic, and independent verification are reported separately for every controlled row. All programs in this audit are root relaxations, not multi-node branch-and-bound runs. The recorded high-water resident memory is cumulative for the sequential process; it cannot support an isolated per-method memory comparison. The raw environment records Python, NumPy, SciPy, platform, and thread settings. Timings are single-run descriptions, not statistically estimated speed advantages.

On LP timeout without usable primal/dual output, zero multipliers give a valid lower bound from the variable rectangle and the operating-optimal reference gives a feasible upper policy. On any other numerical failure the registered runner retains the failure record. Neither a floating infeasibility flag nor a failed sufficient candidate test removes a mathematically feasible policy. The final publication audit requires the actual serialized proof objects to pass; a missing certificate is not silently omitted.

5. COVERAGE AND INHERITED EVIDENCE

Object	Scope of validation
Regret cones and quadratic rate	Analytic proofs in the main article; not machine-checked.
Observable-fiber identity and convergence	Analytic Borel-policy and perturbation arguments; no smearing of atoms.
41 new endpoint objects	Independent reconstruction of full LP lower and deployed-policy upper certificates.
Registered controlled primitives	Independent seeded-generator and parameter check.
Four continuum intervals	Exact independent weighted sums of all required endpoint certificates.
Original 42 maintenance cases	Retained R42 exact gap/savings tabulation; not new R43 optimization runs.
Finest nonlinear object	Inherited R41 full independent replay retained by R42; no new R43 replay claimed.
R42 structural/density cases	Unchanged inherited proofs, results, and verification reports.
Source and publication hashes	Identity and preservation checks only; not economic validity.

The original action-dependent continuous-condition convergence question, difficult approximate-operating-oracle application, tighter nonlinear feasible upper policies, empirical calibration, and a matched marginal local-work ablation are not marked completed. Their scientific records remain unchanged. The new controlled results strengthen the method through explicit structural theorems and computations rather than by weakening the operating constraints or changing the common-policy requirement.

6. REPRODUCTION

From the repository root, set `OPENBLAS_NUM_THREADS=1`, `OMP_NUM_THREADS=1`, and `PYTHONINTMAXSTRDIGITS=0`. Run the registered `replication/run_suite.py`, then `replication/audit.py` and `replication/assemble.py` under the R43 directory. The optional `resume.py` reuses only existing v2 objects that pass a fresh independent endpoint check; it does not change the registered numerical implementation. Compile the four root-level R43 wrappers with `latexmk -pdf -interaction=nonstopmode -halt-on-error`. The publication step checks missing references, overfull horizontal boxes, PDF readability, source hashes, exact endpoint identities, and preservation against the R42 base.

The independent checker uses only Python's standard library and can be run without NumPy, SciPy, an optimizer, or LaTeX. Full numerical regeneration additionally requires NumPy and SciPy. Publication requires the unchanged Econometrica support files, LaTeX, and PyMuPDF. The complete current environment and manifest are stored with the results.